

oil doubled in price, they had to revise their estimate to \$54 (the price of oil is currently, as I am writing these lines, close to \$79 a barrel). It did not dawn on them that it was ludicrous to forecast a second time given that their forecast was off so early and so markedly, that this business of forecasting had to be somehow questioned. And they were looking *twenty-five years* ahead! Nor did it hit them that there was something called an error rate to take into account.*

Forecasting without incorporating an error rate uncovers three fallacies, all arising from the same misconception about the nature of uncertainty.

The first fallacy: *variability matters*. The first error lies in taking a projection too seriously, without heeding its accuracy. Yet, for planning purposes, the accuracy in your forecast matters far more than the forecast itself. I will explain it as follows.

Don't cross a river if it is four feet deep on average. You would take a different set of clothes on your trip to some remote destination if I told you that the temperature was expected to be seventy degrees Fahrenheit, with an expected error rate of forty degrees than if I told you that my margin of error was only five degrees. The policies we need to make decisions on should depend far more on the range of possible outcomes than on the expected final number. I have seen, while working for a bank, how people project cash flows for companies without wrapping them in the thinnest layer of uncertainty. Go to the stockbroker and check on what method they use to forecast sales ten years ahead to “calibrate” their valuation models. Go find out how analysts forecast government deficits. Go to a bank or security-analysis training program and see how they teach trainees to make assumptions; they do not teach you to build an error rate around those assumptions—but their error rate is so large that it is far more significant than the projection itself!

The second fallacy lies in failing to take into account forecast degradation as the projected period lengthens. We do not realize the full extent of the difference between near and far futures. Yet the degradation in such forecasting through time becomes evident through simple introspective examination—without even recourse to scientific papers, which on this topic are suspiciously rare. Consider forecasts, whether economic or technological, made in 1905 for the following quarter of a century. How close to the projections did 1925 turn out to be? For a convincing experience, go read George Orwell's *1984*. Or look at more recent forecasts made in 1975 about the prospects for the new millennium. Many events have taken place and new technologies have appeared that lay outside the forecasters' imaginations; many more that were expected to take place or appear did not do so. Our forecast errors have traditionally been enormous, and there may be no reasons